

Experiment No.	4
Experiment Title	Binary Classification using Linear and Kernel-Based Models
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Github Repository	Click Here

1 Objective

To classify emails as spam or ham using Logistic Regression and Support Vector Machine (SVM) classifiers, and to analyze the effect of hyperparameter tuning (regularization type/strength for Logistic Regression, kernel choice and C/γ for SVM) on classification performance.

2 Problem Statement

Given the Spambase dataset (numerical word/character frequency features extracted from email content), classify each email as spam or ham (binary classification) using a probabilistic linear model (Logistic Regression) and a margin-based kernel model (SVM), and compare their performance before and after hyperparameter tuning.

3 Dataset Description

Dataset Name	spambase.csv (Spambase)
Dataset Source	https://www.kaggle.com/datasets/somesh24/spambase
Number of Samples	4210 (after preprocessing)
Number of Features	57 (word/char frequency + capital-run-length features)
Number of Classes	2 (class: 0 = ham, 1 = spam)
Missing Values	None
Train-Test Split	80% train / 20% test, stratified, random_state=42

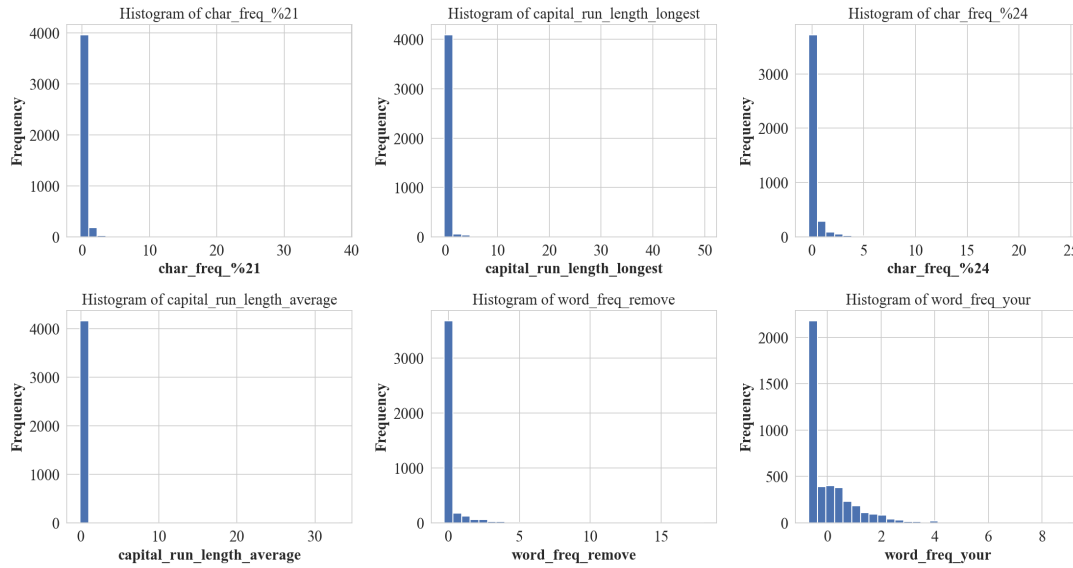


Figure 3: Histogram grid of the top selected features

Inference: Most word-frequency features are heavily right-skewed with a large spike at zero, typical of sparse text-frequency data – this skew is one reason a margin-based, distance-sensitive model like SVM benefits strongly from prior feature standardization.

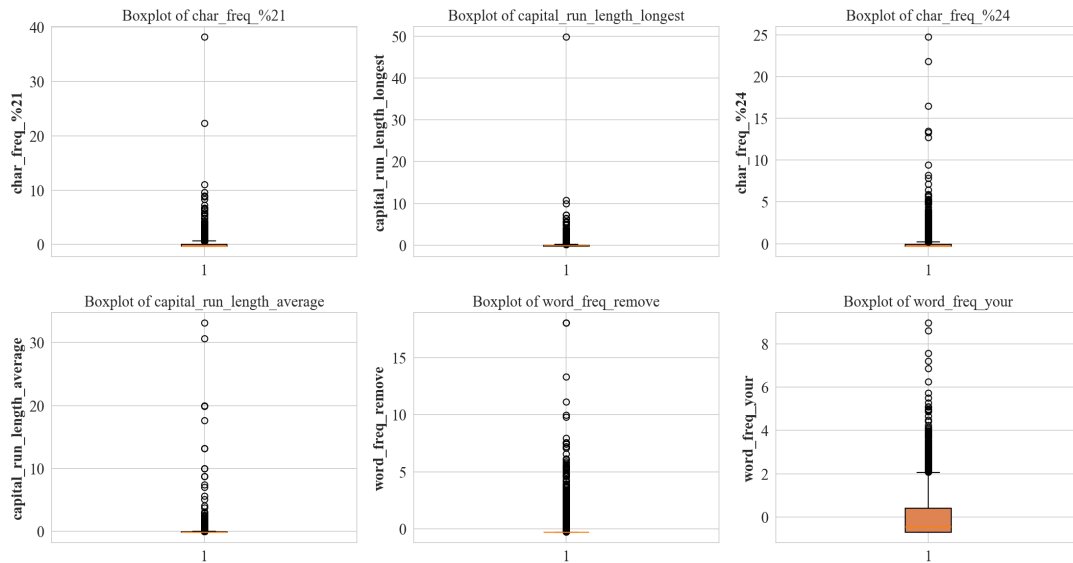


Figure 4: Boxplot grid of the top selected features

Inference: The selected features show a dense cluster near zero with a long tail of outliers, confirming that a small subset of emails repeat spam-indicative words unusually often, which SVM’s margin-maximization is naturally somewhat robust to via support vectors.

5 Data Preprocessing

- No missing values were present after the initial cleaning pipeline (duplicate rows, empty/constant columns removed).

- All numeric features standardized using `StandardScaler` (zero mean, unit variance) – essential for both Logistic Regression convergence and SVM’s distance-based kernels.
- Stratified 80/20 train-test split (`random_state=42`) to preserve the spam/ham ratio.

6 Machine Learning Algorithms

Logistic Regression

Logistic Regression is a probabilistic linear classifier that models the probability of the positive class using the sigmoid function:

$$P(y = 1|x) = \frac{1}{1 + e^{-(w^T x + b)}}$$

It is trained by minimizing log-loss (cross-entropy). L1 regularization (Lasso) encourages sparsity and performs implicit feature selection; L2 regularization (Ridge) penalizes large weights while retaining all features and improves generalization. The parameter C is the inverse regularization strength – small C gives strong regularization (simpler model), large C gives weak regularization (more complex model).

Support Vector Machine (SVM)

SVM finds the optimal hyperplane that maximizes the margin between classes; only the support vectors (points closest to the boundary) determine the decision surface. Kernels extend SVM to non-linear boundaries:

- **Linear:** suitable for linearly separable data.
- **Polynomial:** captures polynomial feature interactions (degree hyperparameter).
- **RBF:** handles complex, non-linear boundaries via a Gaussian similarity measure.
- **Sigmoid:** behaves similarly to a neural-network activation function.

C trades off margin width against misclassification tolerance; γ controls the influence radius of a single training point (relevant for RBF/poly/sigmoid kernels).

7 Experimental Setup

Python Version	
Scikit-Learn Version	
IDE	Jupyter Notebook
Operating System	
Hardware	

8 Hyperparameter Search Space

Parameter	Search Space
Logistic Regression penalty	L1, L2
Logistic Regression C	0.01, 0.1, 1, 10, 100
Logistic Regression solver	liblinear, saga
SVM kernel	Linear, Polynomial, RBF, Sigmoid
SVM C	0.1, 1, 10, 100
SVM γ	scale, auto
SVM degree (Polynomial)	2, 3, 4
Search Method	GridSearchCV, cv=5, scoring='accuracy'

Hyperparameter Tuning Results

Model	Search Method	Best Parameters	Best CV Accuracy
Logistic Regression	Grid	$C = 1$, penalty=l1, solver=liblinear	0.9240
SVM	Grid	kernel=rbf, $C = 1$, γ =auto	0.9249

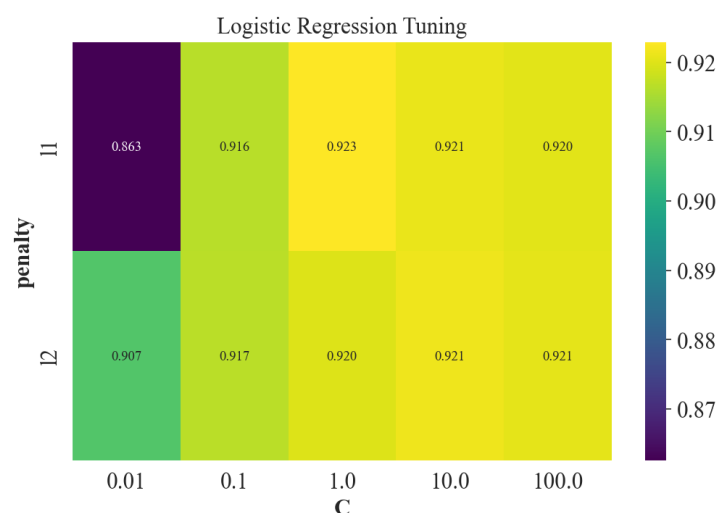


Figure 5: Logistic Regression Tuning – GridSearchCV heatmap (C vs penalty)

Inference: Moderate regularization ($C = 1$) with L1 penalty gives the best cross-validated accuracy; very small C (strong regularization) underfits while very large C (weak regularization) shows only marginal or no improvement, indicating the model is close to its optimal complexity around $C = 1$.

9 Results

Logistic Regression Performance (Baseline)

Metric	Value
Accuracy	0.9347
Precision (weighted)	0.9346
Recall (weighted)	0.9347
F1 Score (weighted)	0.9345
ROC-AUC	0.9767
Training Time (s)	0.1044

SVM Kernel-wise Performance (Baseline, default hyperparameters)

Kernel	Accuracy	F1 Score	Training Time (s)
Linear	0.9311	0.9310	3.1178
Polynomial	0.7660	0.7407	2.9796
RBF	0.9299	0.9296	2.3353
Sigmoid	0.8884	0.8878	1.6734

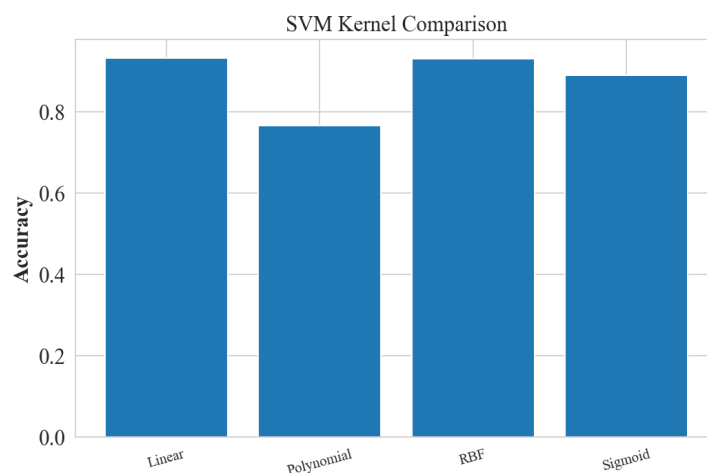


Figure 6: SVM Kernel Comparison (Accuracy)

Inference: The Linear and RBF kernels perform almost identically and clearly outperform Sigmoid and especially Polynomial, suggesting the spam/ham classes are close to linearly separable in this standardized 57-feature space, with RBF adding only marginal non-linear benefit.

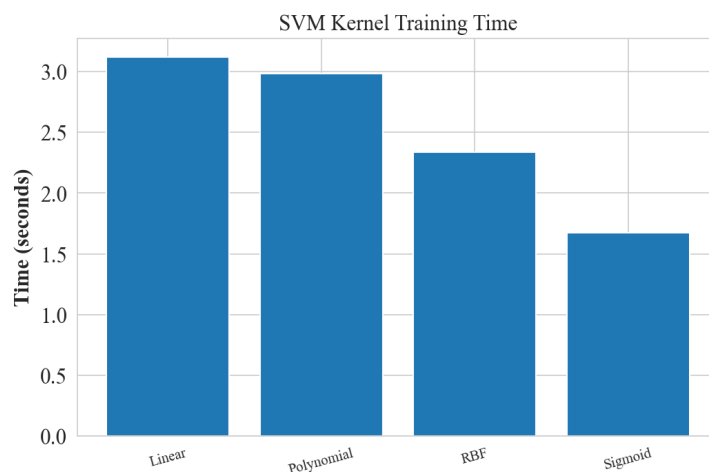


Figure 7: SVM Kernel Training Time

Inference: The Linear kernel takes the longest to train among the four despite its simplicity, since `SVC(kernel='linear')` still solves the full quadratic program rather than using a specialized linear solver, while Sigmoid trains fastest.

Final Tuned Model Comparison

Metric	Logistic Regression (Tuned)	SVM (Tuned, RBF)
Accuracy	0.9371	0.9299
Precision (weighted)	0.9370	0.9300
Recall (weighted)	0.9371	0.9299
F1 Score (weighted)	0.9369	0.9296
ROC-AUC	0.9774	0.9771
Training Time (s)	0.2780	3.3416
Prediction Time (s)	0.0048	0.2347

K-Fold Cross-Validation Results (K=5)

Fold	Logistic Regression (Val Acc)	SVM (Val Acc)
1	0.9169	0.9139
2	0.9258	0.9332
3	0.9243	0.9169
4	0.9198	0.9227
5	0.9331	0.9376
Average	0.9240	0.9249

10 Result Visualization

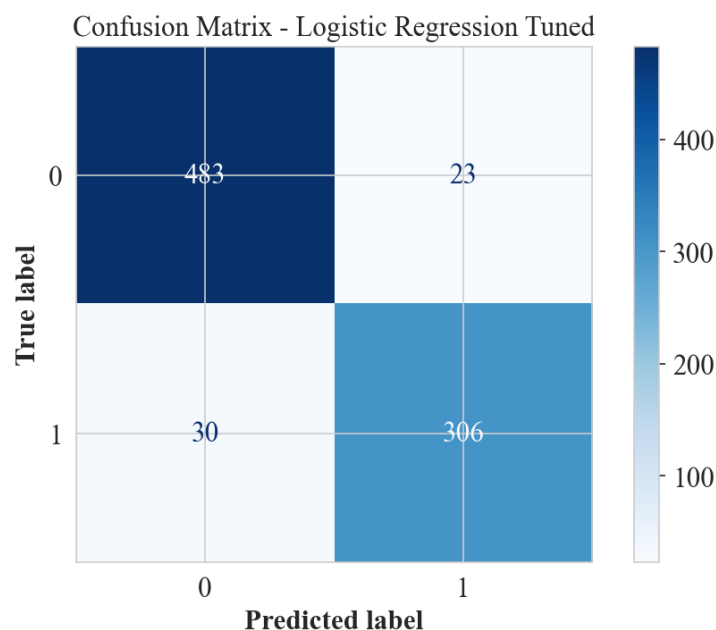


Figure 8: Confusion Matrix – Logistic Regression Tuned

Inference: The tuned Logistic Regression model shows a good balance of errors (23 false positives, 30 false negatives out of 842 test samples), reflecting its strong overall accuracy (93.7%) and confirming L1-regularized tuning improved slightly over the baseline.

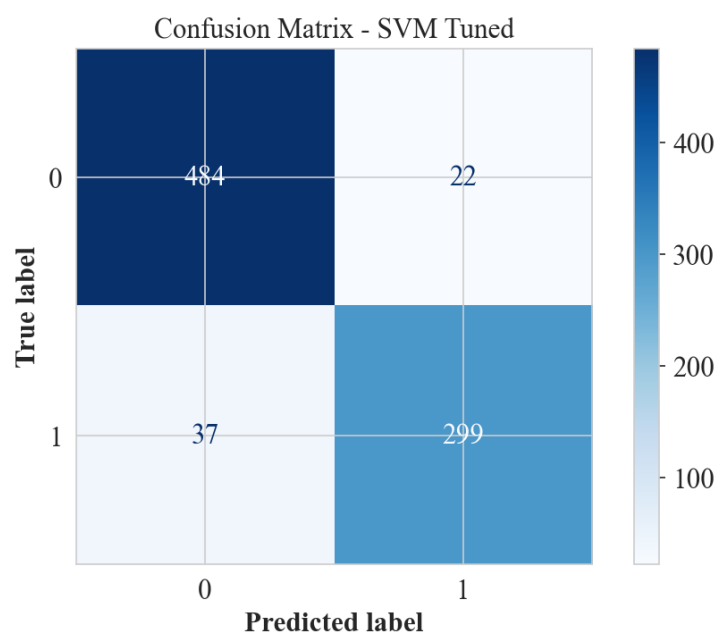


Figure 9: Confusion Matrix – SVM Tuned

Inference: The tuned RBF-kernel SVM shows a marginally higher false-negative count (37) than Logistic Regression, indicating it is slightly more conservative about flagging spam, consistent with its marginally lower recall.

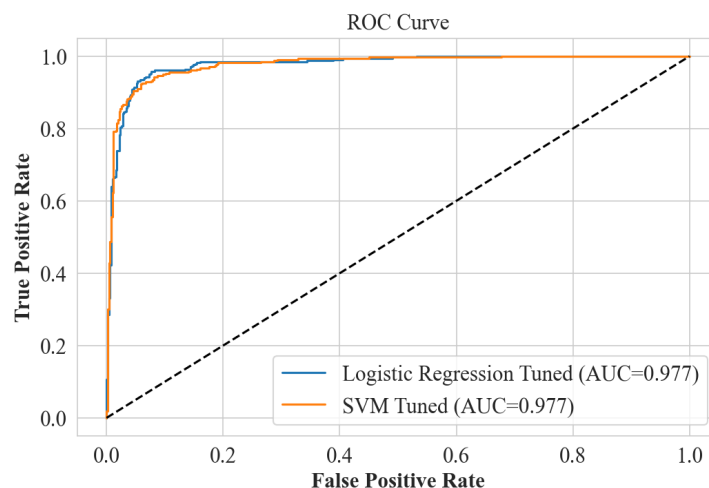


Figure 10: ROC Curve

Inference: Both tuned models achieve nearly identical, very high ROC-AUC (≈ 0.977), with their curves almost overlapping across all thresholds, showing both classifiers rank spam/ham emails with comparable confidence despite their different decision mechanisms.

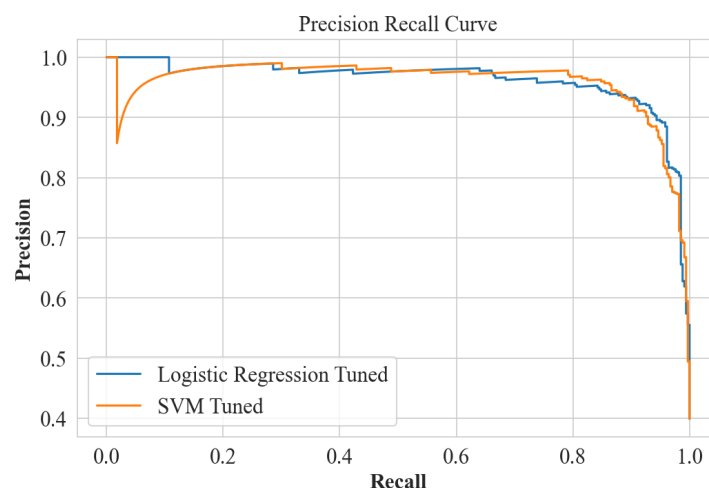


Figure 11: Precision Recall Curve

Inference: Logistic Regression maintains a marginally higher precision across most recall levels than SVM, though the gap is small, reinforcing that the two models are closely matched on this dataset.

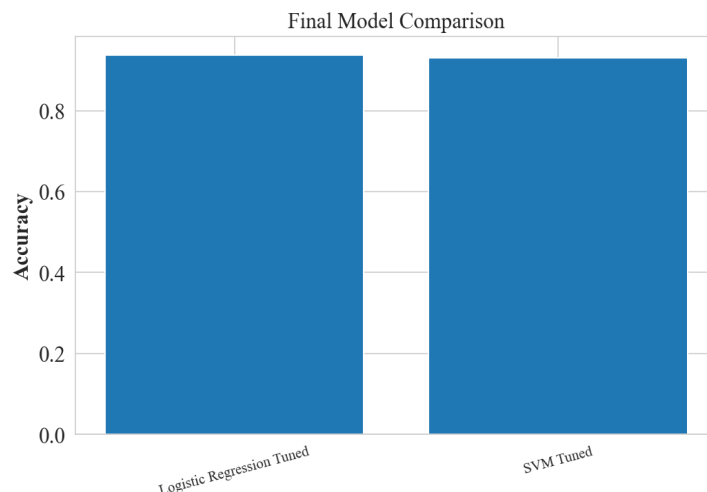


Figure 12: Final Model Comparison (Accuracy)

Inference: Tuned Logistic Regression edges out tuned SVM on test accuracy (93.71% vs 92.99%) while training roughly $12\times$ faster and predicting roughly $49\times$ faster, making it the more practical choice for this dataset unless the marginal accuracy gap becomes critical.

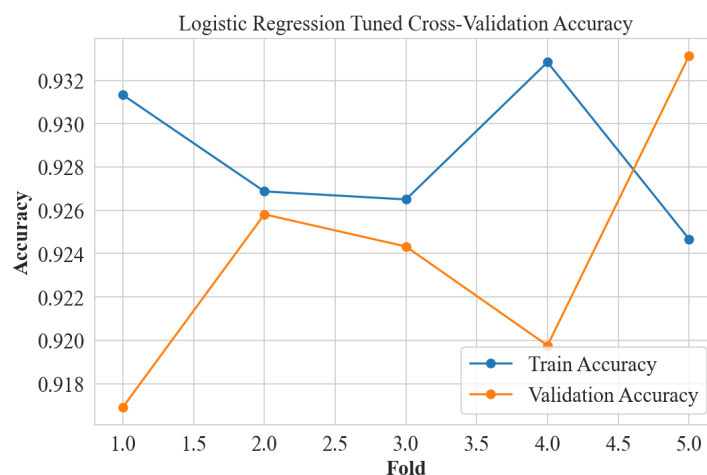


Figure 13: Logistic Regression Tuned – Cross-Validation Accuracy

Inference: Training and validation accuracy remain close across all 5 folds (both around 0.92-0.93), indicating low variance and good generalization with no signs of overfitting after L1 tuning.

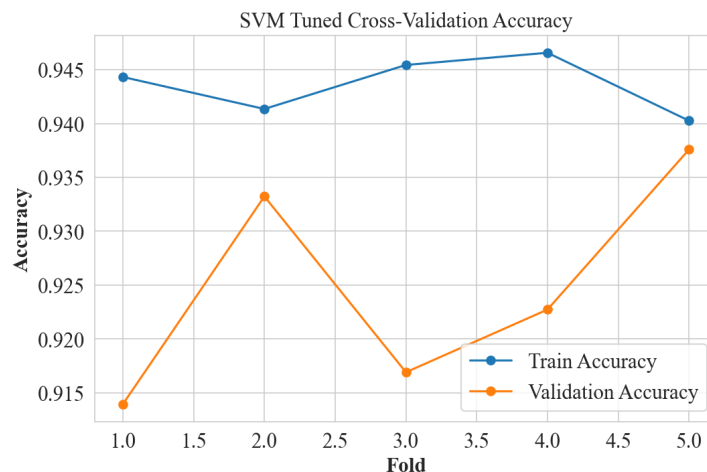


Figure 14: SVM Tuned – Cross-Validation Accuracy

Inference: SVM’s training accuracy is consistently a few points above its validation accuracy across folds, a slightly larger train-validation gap than Logistic Regression, suggesting marginally higher variance despite its comparable average validation accuracy (92.49%).

11 Comparative Analysis

Criterion	Logistic Regression	SVM
Accuracy	0.9371 (higher)	0.9299
Model Complexity	Low	High
Training Time	Low (0.278 s)	High (3.342 s)
Interpretability	High (linear coefficients)	Low (kernel-based)

12 Observations

- **Best-performing classifier:** Tuned Logistic Regression, with the highest test accuracy (93.71%), F1 (0.9369) and ROC-AUC (0.9774), while also being far cheaper to train and predict with.
- **Impact of regularization:** L1 regularization at $C = 1$ improved Logistic Regression’s CV accuracy from the unregularized baseline by inducing sparsity and reducing overfitting risk on the 57-dimensional feature space, without materially hurting accuracy.
- **Kernel behavior in SVM:** Linear and RBF kernels performed almost identically and far better than Polynomial and Sigmoid, indicating the spam/ham classes are close to linearly separable in the standardized feature space – the Polynomial kernel in particular underfit badly (76.6% accuracy) without degree/gamma tuning, while post-tuning RBF (with $\gamma=\text{auto}$) became the best SVM configuration.
- **Bias-variance trade-off:** Logistic Regression showed a smaller train-validation gap across CV folds (lower variance) than SVM, while both had comparably low

bias given their high accuracy; SVM's slightly larger gap suggests marginally more sensitivity to the specific training fold, though not enough to be considered overfitting.

13 Discussion

Both classifiers achieved strong, closely-matched performance (accuracy >93%, ROC-AUC >0.97) on the Spambase dataset after tuning. Logistic Regression's L1-regularized model was the overall best performer and is also the most efficient and interpretable option, since its coefficients directly indicate which word/character frequencies drive spam classification. SVM's RBF kernel matched Linear kernel performance but at a significantly higher computational cost, making it less attractive unless a genuinely more complex decision boundary were needed – which does not appear to be the case for this dataset.

14 Conclusion

Logistic Regression with L1 regularization ($C = 1$) is the recommended classifier for spam detection on the Spambase dataset, offering the best accuracy, fastest training/prediction, and highest interpretability. SVM (RBF kernel, $C = 1$, γ =auto) is a close second in accuracy but is considerably more expensive computationally, making it a reasonable but less efficient alternative. Both models substantially outperform their default/baseline configurations after hyperparameter tuning via GridSearchCV.

15 Learning Outcomes

- Understood the mechanics of probabilistic (Logistic Regression) and margin-based (SVM) classifiers.
- Applied GridSearchCV-based hyperparameter tuning for regularization type/strength and kernel selection.
- Evaluated classification models using accuracy, precision, recall, F1, ROC-AUC and confusion matrices.
- Interpreted experimental results in terms of bias-variance trade-off and model complexity vs interpretability.

16 References

[1] Scikit-learn: Logistic Regression, https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression.

Scikit-learn: Support Vector Machines, <https://scikit-learn.org/stable/modules/svm.html>.

Scikit-learn: Hyperparameter Optimization, https://scikit-learn.org/stable/modules/grid_search.html.

Spambase Dataset, Kaggle, <https://www.kaggle.com/datasets/somesh24/spambase>.

UCI ML Repository – Spambase, <https://archive.ics.uci.edu/dataset/94/spambase>.